Dr. Leon Zaporski

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EDUCATION

University of Cambridge, St. John's College *2019 - 2023*
PhD in Physics, supervised by Prof. Mete Atatüre and Dr. Claire Le Gall
Thesis: 'Coherence and state engineering of an optically active central spin system'

University of Oxford, St. Hilda's College *2015 - 2019*
MMathPhys in Mathematical and Theoretical Physics, Distinction (ranked in the top 10)
BA in Physics, First class honours (Parts A & B)

EXPERIENCE

The Johns Hopkins University (William H. Miller III Department of Physics & Astronomy) 2026 - present
Assistant Professor (tenure-track) *Baltimore, MD*

Massachusetts Institute of Technology (MIT-Harvard Center for Ultracold Atoms) 2023 - 2026
Postdoctoral Associate (Vuletić group) *Cambridge, MA*

Nu Quantum June 2023 - July 2023
Quantum Theorist *Cambridge, UK*

University of Cambridge (Cavendish Laboratory) 2019 - 2023
PhD Candidate (Atatüre group) *Cambridge, UK*

TALKS

- AMO Seminar at National University of Singapore, Singapore, May 2026 (**Invited talk**)
- AMO Seminar at The Johns Hopkins University, USA, March 2026 (**Invited talk**)
- P & A Colloquium at Rice University, USA, March 2026 (**Invited talk**)
- Quantum Science and Engineering Seminar at University of Chicago, USA, February 2026 (**Invited talk**)
- Special Quantum seminar at Princeton University, USA, February 2026 (**Invited talk**)
- Colloquium at Ohio State University, USA, January 2026 (**Invited talk**)
- Cambridge University AMO Seminar, UK, December 2025 (**Invited talk**)
- Quantum Innovation 2025, Osaka, Japan, July - August 2025 (**Invited talk**)
- UConn AMO Seminar, Storrs, USA, May 2025 (**Invited talk**)
- APS March Meeting 2024, Minneapolis, USA, March 2024 (**Invited talk, Session Chair**)
- GaAs Quantum Dots workshop, Oxford, UK, May 2023
- QUANTUMatter 2022, Barcelona, Spain, June 2022
- Quantum 2.0, Boston, USA, June 2022
- CLEO 2022, San Jose, USA, May 2022
- Quantum Dot Day 2022, London, United Kingdom, March 2022

- QIM 2021, Washington, D.C., USA, November 2021
- QD 2020, Munich, Germany, December 2020

AWARDS & SCHOLARSHIPS

- Full EPSRC DTP funding for doctoral training, EP/R513180/1 (2019-2023)
- Captain of the British PLANCKS team (2019)
- Captain of the British PLANCKS team (2018)
- College Prize in Physics, University of Oxford (2018)
- College Prize in Physics, University of Oxford (2017)
- Allen Scholarship, University of Oxford (2016)
- EPSRC Summer Research Grant (2016)
- Finalist of Polish Physics Olympiad (2015)

ACADEMIC REVIEWING

- Reviewer for the Nature Publishing Group (2023-Present)
- Reviewer for the Physical Review (2023-Present)

STUDENT SUPERVISION AND MENTORING

- 2 PhD Students (MIT)
- 2 UROP Students (MIT)
- 2 Master's Students (Cambridge)
- 2 Summer Students (Cambridge)
- 9 Undergraduate Students (Cambridge)

PUBLICATIONS

- **Butterfly Echo Protocol for Axis-Agnostic Heisenberg-Limited Metrology,**
  arXiv:2602.23332 (2026)
- **Quantum-amplified global-phase spectroscopy on an optical clock transition,**
  Nature 646, p. 309–314 (2025)
- **A many-body quantum register for a spin qubit**,
  Nature Physics 21, 368-373 (2025)
- **Tuning the coherent interaction of an electron qubit and a nuclear magnon**,
  PRX 15, 021004 (2025)
- **Many-body singlet prepared by a central spin qubit,**
  PRX Quantum 4, 040343 (2023)
- **Time-crystalline behavior in central-spin models with Heisenberg interactions**,
  PRB 108, 075302 (2023)
- **Ideal refocusing of an optically active spin qubit under strong hyperfine interactions**,
  Nature Nanotechnology 18, p. 257–263 (2023)

- **Optimal purification of a spin ensemble by quantum-algorithmic feedback**,
  PRX 12, 031014 (2022)
- **Witnessing quantum correlations in a nuclear ensemble via an electron spin qubit,**
  Nature Physics 17, p. 1247–1253 (2021)
- **Quantum sensing of a coherent single spin excitation in a nuclear ensemble**,
  Nature Physics 17, p. 585–590 (2021)
- **Superconvergence of topological entropy in the symbolic dynamics of substitution sequences**,
  SciPost Phys. 7, 018 (2019)